Business Performance Analysis

Sales, Profitability, Time Intelligence & Operational Diagnostics

Power BI Analytics + Excel MIS Reporting

Tools: Power BI • Power Query • DAX • MS-Excel
Dataset: Global Super Store Dataset from Kaggle

https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

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From Transaction Data to Management Insights

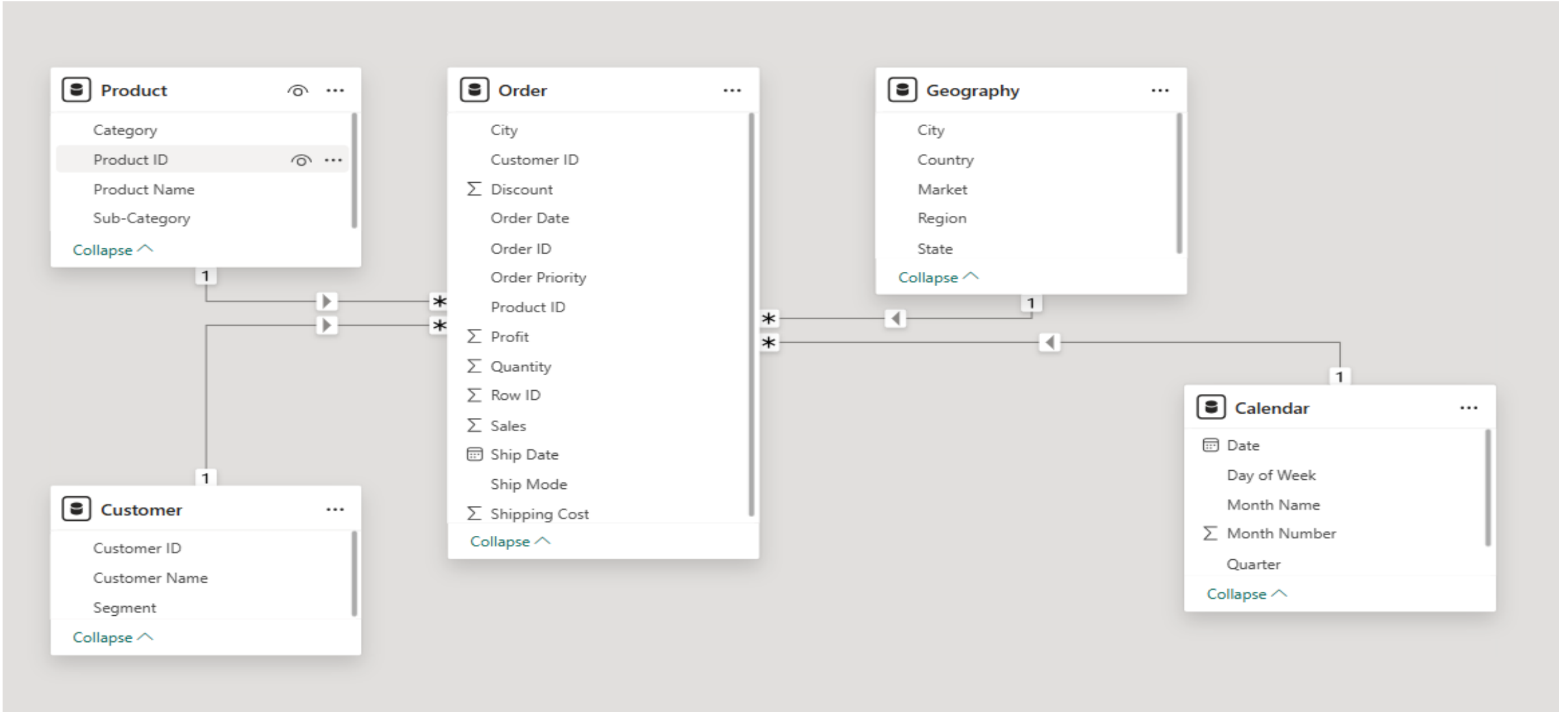
Objective: To develop an integrated reporting solution to monitor:

- Sales & profitability
- Business growth
- Segment & product performance
- Operational costs
- Exceptions requiring attention

Two Reporting Layers

- **Power BI**
  Interactive analysis → *How is performance changing, and where should it be investigated?*
- **Excel MIS**
  Operational monitoring → *Which transactions/areas need attention?*

Data & Analytical Architecture - Star-Schema Based Model



Data Preparation

- **Decomposed the consolidated dataset** into Order, Customer, Product and Geography tables to **reduce data redundancy** and enable **efficient relational analysis**.
- Established **one-to-many relationships** between the Order fact table and dimension tables.
- Created a **Calendar table** for consistent date-based analysis and time intelligence.
- Prepared and standardized fields for reporting and DAX calculations.

Power BI

- DAX measures for Sales, Profit & Margin
- YoY growth calculations
- YTD & Previous-Year YTD
- Rolling 3-month sales
- Interactive filtering & drill-down

Excel MIS

- XLOOKUP for master-data enrichment
- PivotTables / Data Model for summaries
- Dynamic filtering for exception extraction
- PERCENTILE-based exception thresholds

Designed separate analytical and operational reporting layers from the same business data.

Data Visualization
Business Performance Overview

12.64M	51.54%	1.47M	11.61%	178K
Total Sales	YoY Sales Growth %	Total Profit	Profit Margin	Total Quantity

INSIGHTS

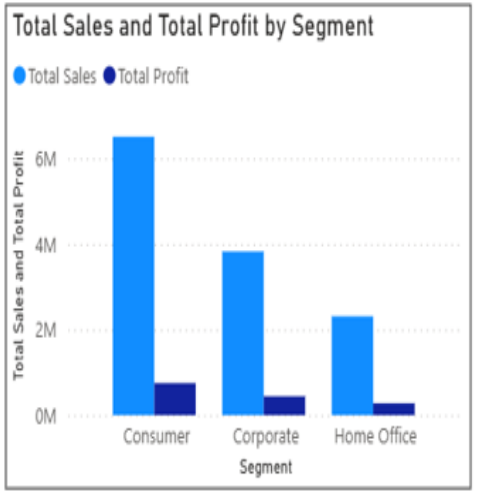
- Technology generated the largest category share of sales (37.53%).
- Central was the highest-selling region, contributing 22.64% of total sales.
- YoY sales growth was 51.54% overall, with September 2012 as the only month recording negative growth.

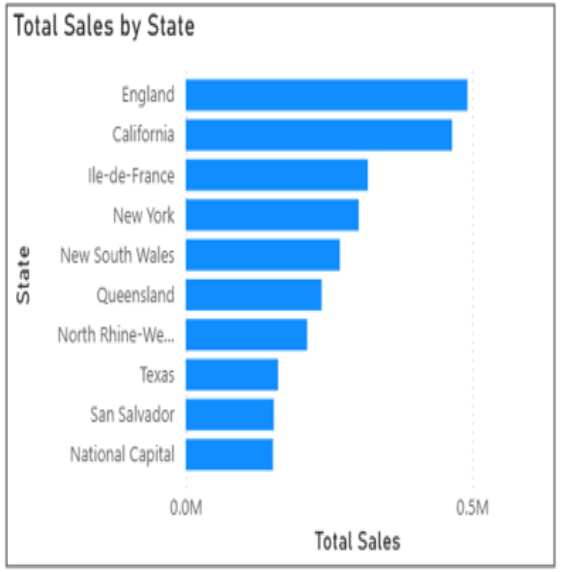


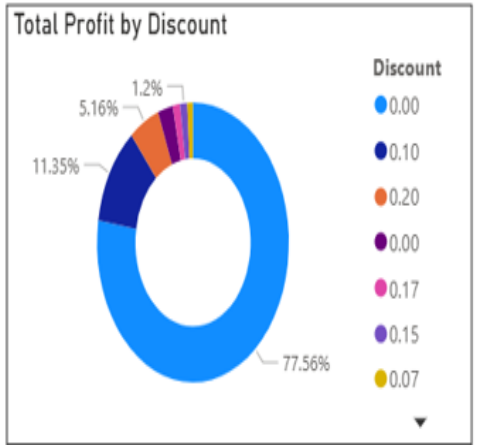
Operational & Segment Analysis

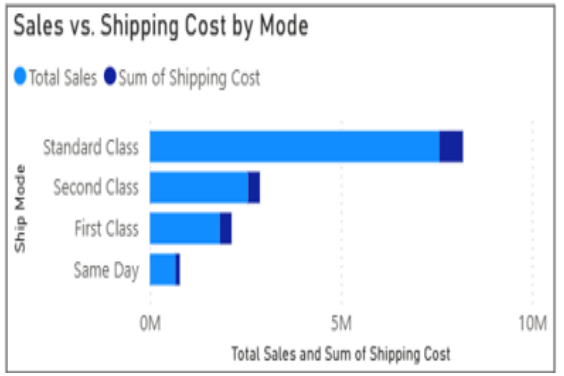
INSIGHTS:

- Consumer generated the highest sales and profit among the three segments, followed by Corporate and Home Office.
- Tables recorded a negative profit of $64.08K and a negative profit margin of 8.46%.
- Standard Class generated the highest sales and also incurred the highest shipping cost among shipping modes.
- England was the highest-selling state, followed by California and Île-de-France.









Category	Total Sales	Total Profit	Profit Margin
Furniture			
Bookcases	14,66,572.24	1,61,924.42	11.04%
Chairs	15,01,681.76	1,40,396.27	9.35%
Furnishings	3,85,578.26	46,967.43	12.18%
Tables	7,57,041.92	-64,083.39	-8.46%
Office Supplies			
Appliances	10,11,064.31	1,41,680.59	14.01%
Art	3,72,091.97	57,953.91	15.58%
Binders	4,61,931.16	72,451.50	15.68%
Envelopes	1,70,904.30	29,601.12	17.32%
Fasteners	83,242.32	11,525.42	13.85%
Labels	73,384.37	15,008.86	20.45%
Paper	2,44,291.72	59,207.68	24.24%
Storage	11,27,085.86	1,08,461.49	9.62%
Supplies	2,43,074.22	22,583.26	9.29%
Technology			
Accessories	7,49,237.02	1,29,626.31	17.30%
Copiers	15,09,436.27	2,58,567.55	17.13%
Machines	7,79,060.07	58,867.87	7.56%
Phones	17,06,824.14	2,16,717.01	12.70%

Growth Pacing & Time Intelligence

INSIGHTS

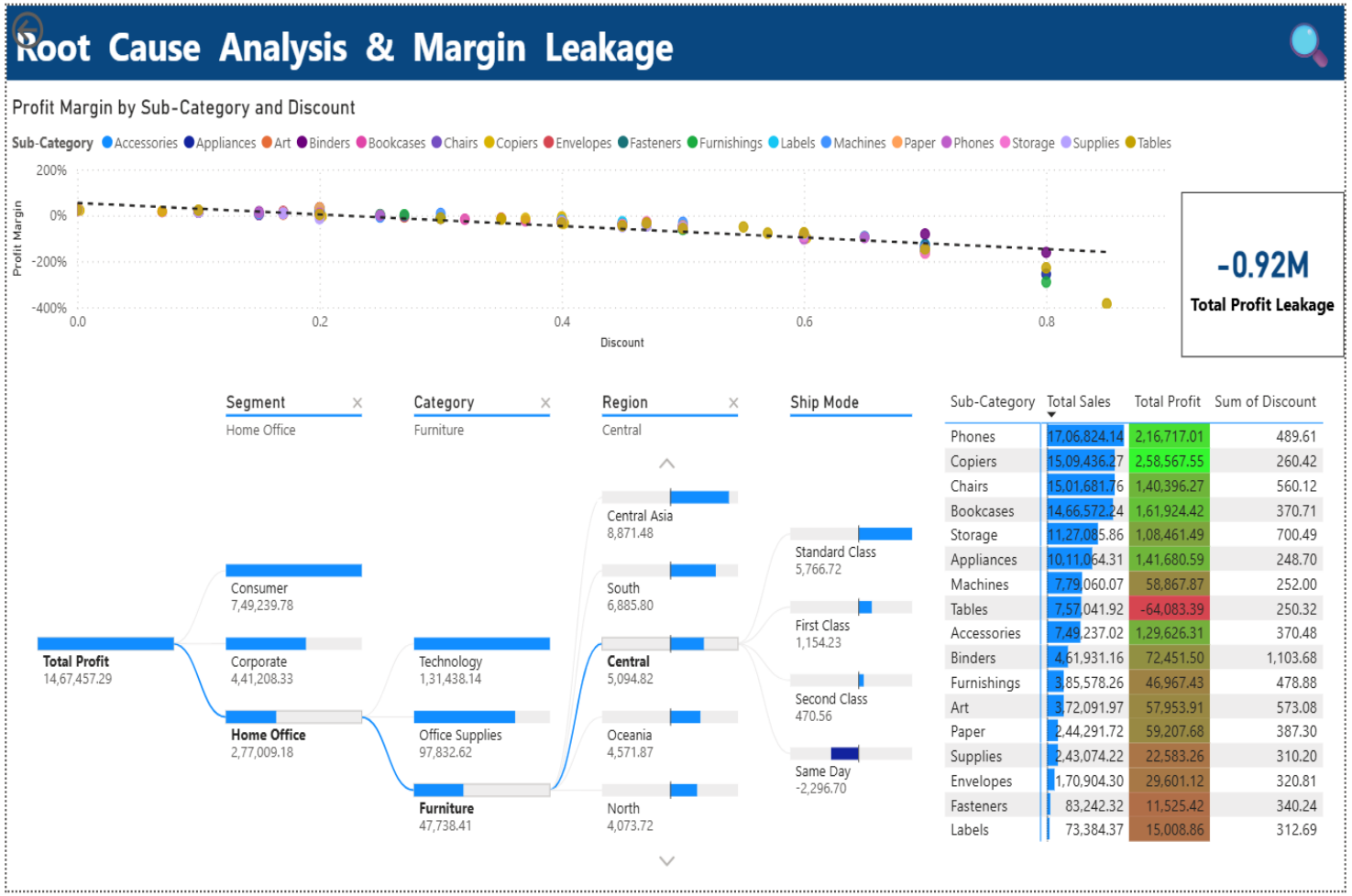
- Monthly YoY sales growth was positive in all months except September 2012, ranging from 46.30% to 56.72% among the positive-growth months.
- October recorded the highest YoY growth at 56.72%, while June recorded the lowest at 46.30%.
- Sales YTD reached $4.30M vs. $3.41M in the previous year, representing approximately 26.25% higher YTD performance.



Diagnostic Analysis

INSIGHTS

- Higher discount levels were associated with weaker profitability in several observations, highlighting potential margin pressure.
- Tables recorded the largest sub-category loss of $64.08K, with a -8.46% profit margin, making it a key profitability concern.
- Total negative-profit transactions amounted to approximately $0.92M, highlighting a significant area of potential profit leakage.



Excel MIS: Performance Monitoring

Region Performance

Region	Total Sales	Total Profit	Total Quantity	Order Count	Profit Margin	Profit Status	Average Discount	Total Shipping Cost	AOV	Sales Contribution %
East	665017.21	96506.30	10451	1401	14.51%	Low	13.65%	71108.30	474.6732	5.26%
Oceania	1101009.71	120137.42	12813	1753	10.91%	Low	15.18%	120946.77	628.0717	8.71%
Central	2862200.29	317127.36	42271	5316	11.08%	High	13.87%	300923.99	538.4124	22.64%
Africa	758111.42	82612.13	10277	2180	10.90%	Low	16.12%	85464.39	347.7575	6.00%
West	714375.12	116028.20	11771	1563	16.24%	Low	9.93%	74895.46	457.0538	5.65%
South	1599785.87	130833.85	25244	3283	8.18%	Low	15.20%	168689.30	487.2939	12.65%
Central Asia	755418.99	132664.51	7833	1037	17.56%	Low	6.99%	78052.04	728.4658	5.98%
EMEA	800053.33	40124.51	11463	2440	5.02%	Low	19.84%	87548.31	327.8907	6.33%
North Asia	848269.55	165595.43	8842	1149	19.52%	Medium	4.85%	95027.29	738.2677	6.71%
North	1271207.41	197587.82	18438	2414	15.54%	Medium	9.83%	134144.16	526.5979	10.06%
Caribbean	327175.15	35477.88	6375	865	10.84%	Low	13.50%	35854.83	378.2372	2.59%
Southeast Asia	882082.82	17702.97	11798	1512	2.01%	Low	27.19%	93565.30	583.3881	6.98%
Canada	57795.04	15058.93	736	182	26.06%	Low	0.06%	6600.55	317.5551	0.46%
Total	12642501.91	1467457.29	178312.00	25095	11.61%	High	12.79%	1352820.69	503.79	1.00

- Central generated the highest regional sales at approximately $2.86M.
- Consumer was the largest segment by sales at approximately $6.51M.

Segment Performance

Segment	Total Sales	Total Profit	Total Quantity	Order Count	Profit Margin	Average Discount	Total Shipping Cost	AOV	Sales Contribution %	Profit Contribution %	Profit Status
Consumer	6507949.42	749239.78	92157	13104	11.51%	29.06%	697300.64	496.64	51.48%	51.06%	High
Corporate	3824697.52	441208.33	53565	7673	11.54%	28.74%	410474.46	498.46	30.25%	30.07%	Medium
Home Office	2309854.97	277009.18	32590	4687	11.99%	28.09%	245045.59	492.82	18.27%	18.88%	Low
Total	12642501.91	1467457.29	178312.00	25464	11.61%	28.78%	1352820.69	496.49	100.00%	100.00%	Medium

Excel MIS: Product Performance

A	B	C	D	E	F	G	H	I	J	K	L	M
Product Performance												
Sub-Category	Category	Total Sales	Total Profit	Total Quantity	Order Count	Profit Margin	Average Discount	Total Shipping Cost	AOV	Sales Contribution %	Profit Contribution %	Profit Status
Accessories	Technology	749237.02	129626.31	10946	2889	17.30%	12.82%	83513.58	259.34	5.93%	8.83%	High
Chairs	Furniture	1501681.76	140396.27	12336	3187	9.35%	17.58%	164229.85	471.19	11.88%	9.57%	High
Phones	Technology	1706824.14	216717.01	11870	3133	12.70%	15.63%	184902.77	544.79	13.50%	14.77%	High
Copiers	Technology	1509436.27	258567.55	7454	2120	17.13%	12.28%	159496.49	712.00	11.94%	17.62%	High
Tables	Furniture	757041.92	-64083.39	3083	836	-8.46%	29.94%	79861.46	905.55	5.99%	-4.37%	Loss
Binders	Office Supplies	461931.16	72451.50	21433	5394	15.68%	20.46%	48184.08	85.64	3.65%	4.94%	Medium
Supplies	Office Supplies	243074.22	22583.26	8543	2281	9.29%	13.60%	24812.05	106.56	1.92%	1.54%	Low
Appliances	Office Supplies	1011064.31	141680.59	6078	1686	14.01%	14.75%	108300.78	599.68	8.00%	9.65%	High
Machines	Technology	779060.07	58867.87	4906	1422	7.56%	17.72%	79135.9	547.86	6.16%	4.01%	Medium
Bookcases	Furniture	1466572.24	161924.42	8310	2284	11.04%	16.23%	155482.23	642.11	11.60%	11.03%	High
Storage	Office Supplies	1127085.86	108461.49	16917	4534	9.62%	15.45%	120546.46	248.59	8.92%	7.39%	Medium
Furnishings	Furniture	385578.26	46967.43	11225	2965	12.18%	16.15%	40747.12	130.04	3.05%	3.20%	Low
Art	Office Supplies	372091.97	57953.91	16301	4366	15.58%	13.13%	41287.47	85.22	2.94%	3.95%	Medium
Paper	Office Supplies	244291.72	59207.68	12822	3234	24.24%	11.98%	26661.29	75.54	1.93%	4.03%	Medium
Envelopes	Office Supplies	170904.30	29601.12	8380	2310	17.32%	13.89%	18547.78	73.98	1.35%	2.02%	Low
Fasteners	Office Supplies	83242.32	11525.42	8390	2304	13.85%	14.77%	9053.59	36.13	0.66%	0.79%	Low
Labels	Office Supplies	73384.37	15008.86	9318	2457	20.45%	12.73%	8057.79	29.87	0.58%	1.02%	Low
Total		12642501.91	1467457.29	178312.00	25035	11.61%		1352820.69	504.99	100.00%	100.00%	

- Copiers was the most profitable sub-category, generating $258.57K profit and contributing 17.62% of total profit.
- Tables was the only loss-making sub-category, recording a $64.08K loss with a -8.46% profit margin.

Excel MIS: Exception-Based Monitoring

Exception Report

Order ID	Order Date	Region	Segmer	Category	Sub-Ca	Sales	Profit	Profit Mar	Discou	Shippin	Order Pi	Exception Type
CA-2012-12489	31-07-2012	East	Consumer	Technology	Accessories	2309.7	762.18	33.00%	0.00%	933.57	Critical	High Shipping Cost
IN-2013-77878	05-02-2013	Oceania	Corporate	Furniture	Chairs	3709.4	-288.77	-7.78%	10.00%	923.63	Critical	Loss-Making
IN-2013-71249	17-10-2013	Oceania	Consumer	Technology	Phones	5175.2	919.97	17.78%	10.00%	915.49	Medium	High Shipping Cost
ES-2013-15793	28-01-2013	Central	Home Office	Technology	Phones	2892.5	-96.54	-3.34%	10.00%	910.16	Medium	Loss-Making
SG-2013-4320	05-11-2013	Africa	Consumer	Technology	Copiers	2833	311.52	11.00%	0.00%	903.04	Critical	High Shipping Cost
IN-2013-42360	28-06-2013	Oceania	Corporate	Technology	Phones	2862.7	763.28	26.66%	10.00%	897.35	Critical	High Shipping Cost
IN-2011-81826	07-11-2011	Oceania	Consumer	Furniture	Chairs	1822.1	564.84	31.00%	0.00%	894.77	Critical	High Shipping Cost
IN-2012-86369	14-04-2012	Oceania	Consumer	Furniture	Tables	5244.8	996.48	19.00%	0.00%	878.38	High	High Shipping Cost
CA-2014-13590	14-10-2014	West	Corporate	Office Supplie	Binders	5084	1906.5	37.50%	20.00%	867.69	Low	High Discount
CA-2012-11663	28-01-2012	South	Consumer	Furniture	Tables	4297.6	-1862.3	-43.33%	40.00%	865.74	Critical	Loss-Making
CA-2011-10298	05-04-2011	South	Corporate	Office Supplie	Supplies	4164.1	83.281	2.00%	0.00%	846.54	High	Low Margin
ID-2012-28402	19-04-2012	Central Asia	Corporate	Furniture	Tables	4626.2	647.55	14.00%	0.00%	835.57	High	High Shipping Cost
SA-2011-1830	27-12-2011	EMEA	Consumer	Technology	Phones	2617	1151.4	44.00%	0.00%	832.41	Critical	High Shipping Cost
MX-2012-13001	13-11-2012	South	Home Office	Furniture	Chairs	2221.8	622.02	28.00%	0.00%	810.25	Critical	High Shipping Cost
IN-2013-73951	06-06-2013	North Asia	Consumer	Office Supplie	Appliances	3701.5	1036.1	27.99%	0.00%	804.54	Critical	High Shipping Cost
ES-2014-50999	31-07-2014	Central	Corporate	Office Supplie	Appliances	1869.6	186.95	10.00%	10.00%	801.66	Critical	High Shipping Cost
CA-2014-14356	03-11-2014	South	Corporate	Technology	Accessories	2249.9	517.48	23.00%	0.00%	780.7	Critical	High Shipping Cost
ES-2014-165177	08-09-2014	South	Corporate	Office Supplie	Appliances	7958.6	3979.1	50.00%	0.00%	778.32	Low	High Shipping Cost
IN-2014-11763	31-01-2014	Oceania	Corporate	Technology	Copiers	2565.6	28.404	1.11%	10.00%	766.93	Critical	Low Margin
TZ-2014-8190	05-12-2014	Africa	Consumer	Office Supplie	Appliances	3409.7	818.28	24.00%	0.00%	763.38	High	High Shipping Cost
PL-2012-7820	08-08-2012	EMEA	Corporate	Furniture	Tables	1977.7	276.84	14.00%	0.00%	759.47	Critical	High Shipping Cost
CA-2011-154627	29-10-2011	Central	Consumer	Technology	Phones	2736	341.99	12.50%	20.00%	752.51	High	High Discount
IN-2011-44803	02-05-2011	North Asia	Corporate	Furniture	Chairs	2754	358.02	13.00%	0.00%	752.47	Critical	High Shipping Cost
ES-2013-28605	27-02-2013	North	Home Office	Office Supplie	Appliances	5273.7	1898.4	36.00%	0.00%	730.91	High	High Shipping Cost
US-2014-13319	31-07-2014	North	Consumer	Technology	Phones	1713.8	445.52	26.00%	0.00%	728.97	Critical	High Shipping Cost
MX-2014-16530	05-09-2014	Central	Consumer	Furniture	Tables	2106.5	526.5	24.99%	20.00%	728.39	Critical	High Discount
IN-2011-10286	17-12-2011	North Asia	Corporate	Furniture	Tables	1715.2	720.36	42.00%	0.00%	725.57	Critical	High Shipping Cost
ES-2011-46937	14-03-2011	Central	Consumer	Office Supplie	Appliances	3069.7	1364.2	44.44%	10.00%	725.34	Critical	High Shipping Cost
CA-2013-15901	11-03-2013	West	Home Office	Technology	Phones	4158.9	363.9	8.75%	20.00%	714.66	High	High Discount
IN-2012-44810	25-02-2012	Central Asia	Consumer	Furniture	Chairs	1878.7	582.36	31.00%	0.00%	704.08	Critical	High Shipping Cost
US-2011-12877	28-12-2011	Caribbean	Corporate	Technology	Phones	1696.6	-148.46	-8.75%	20.00%	704.06	Critical	Loss-Making
ES-2012-58702	17-07-2012	Central	Corporate	Technology	Machines	2402.9	763.16	31.76%	15.00%	699.55	Critical	High Shipping Cost
CA-2012-13973	15-10-2012	Central	Consumer	Furniture	Chairs	2453.4	-350.49	-14.29%	30.00%	690.42	High	Loss-Making
IN-2011-28087	03-11-2011	Oceania	Corporate	Office Supplie	Appliances	2526.9	561.48	22.22%	10.00%	689.8	Critical	High Shipping Cost
CA-2011-16849	12-12-2011	West	Consumer	Furniture	Tables	3610.8	135.41	3.75%	20.00%	683.12	High	High Discount + Low Margin
CG-2011-8610	14-09-2011	Africa	Corporate	Technology	Phones	3817.3	1068.7	28.00%	0.00%	678.15	High	High Shipping Cost
CA-2011-16076	14-09-2011	East	Consumer	Technology	Machines	2800	1372	49.00%	0.00%	675.15	High	High Shipping Cost
US-2014-16811	05-11-2014	South	Corporate	Technology	Machines	8000	-3840	-48.00%	50.00%	674.82	High	Loss-Making
ES-2014-26372	14-01-2014	North	Consumer	Technology	Copiers	4141	1697.7	41.00%	0.00%	668.96	High	High Shipping Cost
IN-2011-61302	10-01-2011	Oceania	Consumer	Technology	Phones	2875.1	511.1	17.78%	10.00%	665.27	Medium	High Shipping Cost

Exception Identification Logic

Exception Type	Threshold / Logic
Loss-Making	Profit < 0
High Discount	Discount ≥ **75th Percentile (20%)**
Low Margin	Positive Profit Margin ≤ **10th Percentile (5.96%)**
High Shipping Cost	Shipping Cost ≥ **99th Percentile (₹286.75)**
High Discount + Low Margin	**Both High Discount and Low Margin thresholds satisfied**

Project Outcomes & Key Takeaways

Power BI

- Analytical & Interactive
- KPI monitoring
- Time intelligence
- Segmentation
- Geographic analysis
- Profitability diagnostics

Excel MIS

- Operational & Exception-Based
- Structured MIS reports
- Performance classification
- Transaction-level exception reporting
- Conditional formatting
- Filter-based exception identification.

Business Outcome:

Integrated analytical and operational reporting to monitor performance, identify profitability issues and highlight transactions requiring attention.